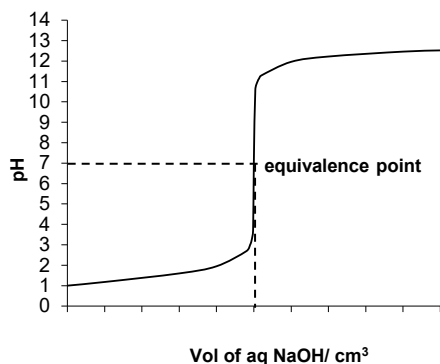
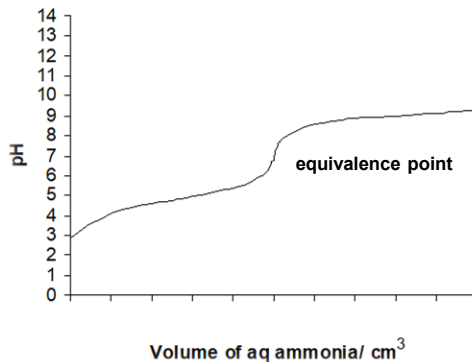


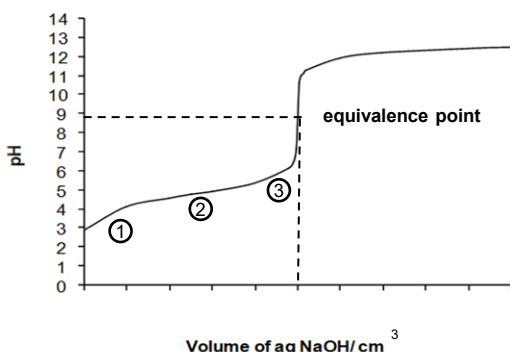
Strong Acid – Strong Base e.g. $\text{HCl} / - \text{NaOH}$



Weak Acid – Weak Base e.g. $\text{CH}_3\text{COOH} - \text{NH}_3$



Weak Acid – Strong Base e.g. $\text{CH}_3\text{COOH} - \text{NaOH}$



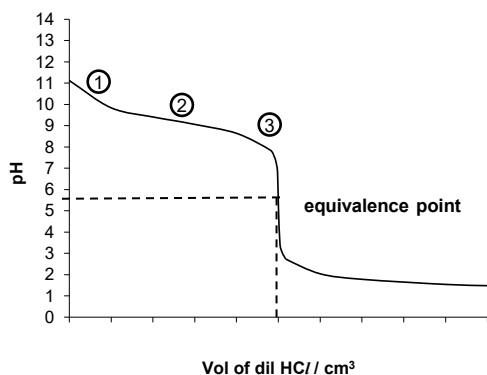
- Why is the pH at equivalence point more than 7?
At equivalence point, the solution contains
..... hydrolyses partially in water to produce
equation:
- When is pH of solution = pK_a of CH_3COOH ?
- Indicate the pK_a of CH_3COOH by drawing construction lines on the graph.
- What are the **major** species present at the indicated positions?

① :

② :

③ :

Weak Base – Strong Acid e.g. $\text{NH}_3 - \text{HCl}$



- Why is the pH at equivalence point less than 7?
At equivalence point, the solution contains
..... hydrolyses partially in water to produce
equation:
- By considering the species present, explain why the slope of the graph only changes gradually at position (2) when more HCl is being added.

At position (2), the solution is

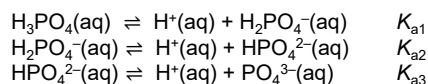
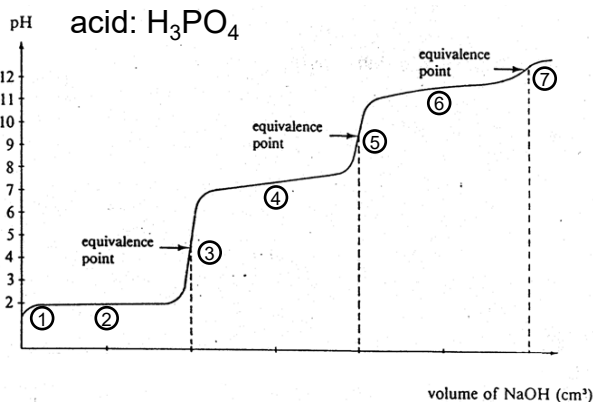
When HCl is being added,

.....

.....

equation:

Polyprotic acid titration curve



- Indicate pK_{a1} , pK_{a2} & pK_{a3} by drawing construction lines on the graph.
- What are the **major** species present at the indicated positions?

① :

② :

③ :

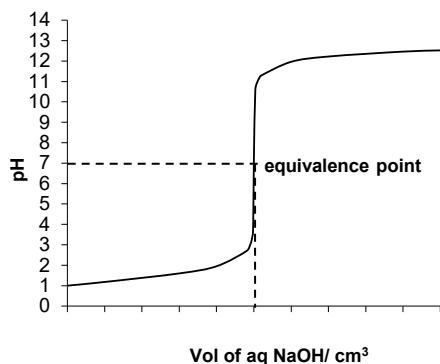
④ :

⑤ :

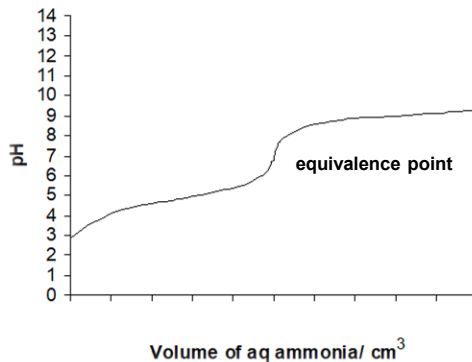
⑥ :

⑦ :

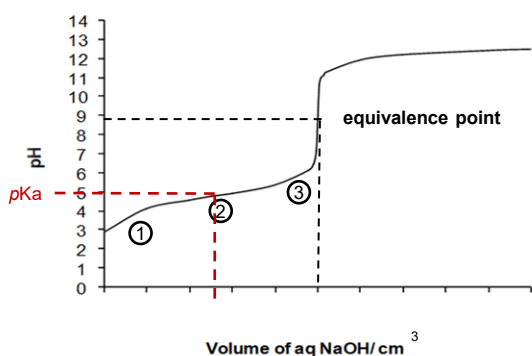
Strong Acid – Strong Base e.g. $\text{HCl} - \text{NaOH}$



Weak Acid – Weak Base e.g. $\text{CH}_3\text{COOH} - \text{NH}_3$

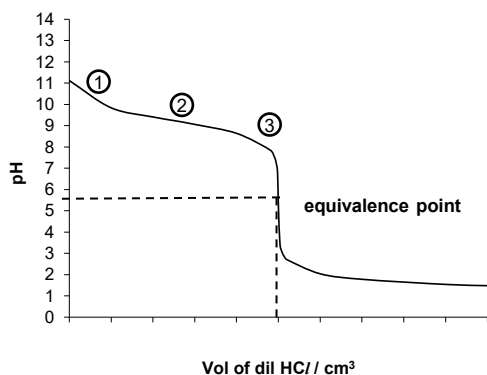


Weak Acid – Strong Base e.g. $\text{CH}_3\text{COOH} - \text{NaOH}$



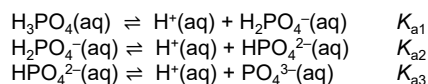
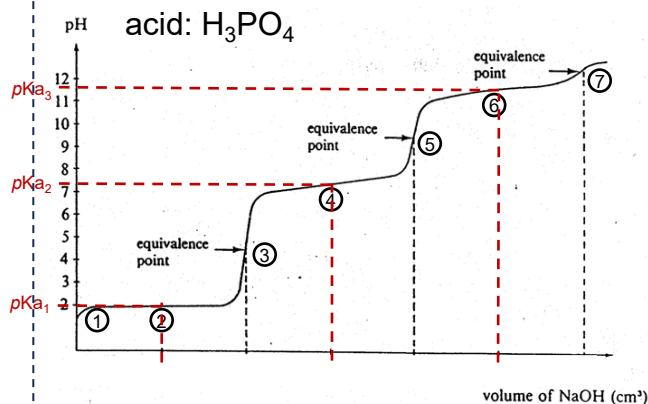
- Why is the pH at equivalence point more than 7?
At equivalence point, the solution contains CH_3COONa
 CH_3COO^- hydrolyses partially in water to produce OH^- ions
equation: $\text{CH}_3\text{COO}^- + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COOH} + \text{OH}^-$
- When is pH of solution = pK_a of CH_3COOH ?
At (2), when $\text{CH}_3\text{COO}^- : \text{CH}_3\text{COOH} = 1:1$
The solution is an acidic buffer is at max buffer capacity
- Indicate the pK_a of CH_3COOH by drawing construction lines on the graph.
- What are the **major** species present at the indicated positions?
① : CH_3COOH
② : $\text{CH}_3\text{COOH}, \text{CH}_3\text{COO}^-$; 1:1
③ : CH_3COO^-

Weak Base – Strong Acid e.g. $\text{NH}_3 - \text{HCl}$



- Why is the pH at equivalence point less than 7?
At equivalence point, the solution contains NH_4Cl
 NH_4^+ hydrolyses partially in water to produce H^+ ions
equation: $\text{NH}_4^+ \rightleftharpoons \text{NH}_3 + \text{H}^+$
- By considering the species present, explain why the slope of the graph only changes gradually at position (2) when more HCl is being added.
At position (2), the solution is $\text{a buffer containing } \text{NH}_4^+ \text{ and } \text{NH}_3$
When HCl is being added, $\text{by LCP, the equilibrium position of } \text{NH}_4^+ \rightleftharpoons \text{NH}_3 + \text{H}^+ \text{ shifts to the left, to partially offset the increase in } [\text{H}^+], \text{ resulting in a small decrease in pH for the addition of HCl}$
equation: $\text{NH}_3 + \text{H}^+ \rightarrow \text{NH}_4^+$

Polyprotic acid titration curve



- Indicate pK_{a1} , pK_{a2} & pK_{a3} by drawing construction lines on the graph.
- What are the **major** species present at the indicated positions?
① : H_3PO_4
② : $\text{H}_3\text{PO}_4, \text{H}_2\text{PO}_4^-$; 1:1
③ : H_2PO_4^-
④ : $\text{H}_2\text{PO}_4^-, \text{HPO}_4^{2-}$; 1:1
⑤ : HPO_4^{2-}
⑥ : $\text{HPO}_4^{2-}, \text{PO}_4^{3-}$; 1:1
⑦ : PO_4^{3-}